

Semester:	3rd [cite: 1]	Subject Code:	DI03016061 [cite: 1]
Subject Name:	Operating Systems [cite: 1]	Last Date:	1/10/2026
Assignment:	Unit - 4 (File Management) [cite: 1]		

Assignment Questions & Bloom's Taxonomy Mapping

Q.No.	Question Description	RBT Level
1.	Define the File concept in operating systems and describe its primary properties [cite: 1].	Remember (R)
2.	List and explain various file attributes and file types supported by an operating system [cite: 1].	Understand (U)
3.	Explain different file access methods, specifically focusing on Sequential access and Direct access [cite: 1].	Understand (U)
4.	Describe various file allocation methods (Contiguous, Linked, and Indexed allocation) used for disk space management [cite: 1].	Understand (U)
5.	Explain the concept of file sharing in multi-user operating systems and discuss the mechanisms used for it [cite: 1].	Understand (U)
6.	Discuss file protection mechanisms. How are access control lists (ACLs) utilized to secure files against unauthorized access [cite: 1]?	Understand (U)
7.	A file system uses indexed allocation with 10 direct blocks, 1 single-indirect block, 1 double-indirect block, and 1 triple-indirect block. If a disk block pointer requires 4 bytes and the block size is 1 KB, calculate the maximum possible file size [cite: 1].	Application (A)

Note: Submit the handwritten or neatly typed assignment on or before the specified last date. Ensure proper diagrams and block structures are drawn wherever applicable.